

Frames in the Span: Exact, Label-Free Recovery of Long-Form Speech from Federated ASR Updates

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Abstract

Federated speech recognition sends model updates to a server and keeps the audio on the device. We show that for acoustic models whose first layer applies one linear map to overlapping windows of feature frames, a single update contains the features. The rows of the first-layer gradient span the windows, the overlap turns membership in that span into a linear system, and a banded solve returns the utterance without transcript, length, loss or optimisation. The system is uniquely solvable up to $(k-s)F$ windows for kernel k , stride s and F features. On LibriSpeech the closed form recovers 98.5% of utterances up to 6.2 s from DeepSpeech and 94.8% up to 9.6 s from a Transformer-CTC front end; a sequential decoder over deeper layers reaches 35 s. Synthesised audio is transcribed at 3.8% WER and the speaker identified in 98% of cases. Local SGD, batching, clipping, dropout and training do not prevent the attack; local Adam, pruning and sufficient noise do.

Index Terms: federated learning, speech recognition, privacy, gradient inversion, CTC

1. Introduction

Federated learning (FL) [1] is attractive for automatic speech recognition (ASR) because the audio stays on the device [2, 3, 4]. What leaves the device is a model update, and gradient inversion [5, 6] asks how much of the input an update reveals.

For speech the evidence comes from optimisation attacks. Dang et al. [7] search for DeepSpeech [8] input features whose gradient matches the observed one, with the transcript and length of short utterances given, and identify the speaker in 34% of cases; a dropout rate of 0.2 reduces this to 0%. Their label attack [9] recovers bags of word pieces. Later work matches gradients of small classifiers on one-second clips [10, 11], scales to long audio by reconstructing segments of a CNN classifier’s input [12], or descends through the CTC loss [13] on DeepSpeech [14]. Speaker information also leaks from adapted weights without reconstruction [15]. Optimisation attacks are slow, assume side information, and fail under dropout, local training or long inputs; such failures support the view that gradient leakage is fragile in realistic FL [16, 17].

A second line of work is algebraic. The gradient of a linear layer is a sum of outer products, so its row space is the span of the layer’s inputs. DAGER [18] tests discrete tokens against this span, SPEAR [19] separates small image batches with it, and TIGER [20] optimises embeddings towards it. None of these addresses speech.

We observe that speech front ends make the span unusually easy to read. The frames of one utterance act as a batch, and every frame is seen through a window that overlaps its neighbours: spliced context in DeepSpeech and TDNN-style models,

a strided convolution in Transformer encoders. The overlap is what converts span membership into a linear system in the features. We contribute (i) a closed-form attack with a capacity law verified on four front ends, (ii) a sequential decoder for utterances beyond that capacity, (iii) an evaluation under the conditions of federated training, including trained models, and (iv) a map of which architectures and defences stop the attack.

2. The span leak

2.1. Setting

A client holds features $X = (x_1, \dots, x_T)$, $x_t \in \mathbb{R}^F$. The first layer reads windows $c_p(X) = [x_{ps-a}, \dots, x_{ps-a+k-1}] \in \mathbb{R}^{kF}$, $p = 1, \dots, P$, zero outside $1..T$, and computes $Wc_p + b$ with $W \in \mathbb{R}^{H \times kF}$. DeepSpeech-1 splices c frames of context on each side ($k = 2c + 1$, $s = 1$). The Transformer encoder of the FL benchmark in [4] uses a 1-D convolution with $k = 7$, $s = 3$ on $F = 80$ log-mel bins. The server is honest but curious: it knows the model it sent and receives the update of one client.

2.2. Span and length

For any loss L and any later layers, with $g_p = \partial L / \partial (Wc_p + b)$,

$$\nabla_W L = \sum_{p=1}^P g_p c_p^\top, \quad \nabla_b L = \sum_{p=1}^P g_p. \quad (1)$$

The rows of $M = [\nabla_W L \mid \nabla_b L]$ lie in $\mathcal{S} = \text{span}\{[c_p, 1]\}_p$, and the row space equals $\mathcal{S}$ when the g_p are linearly independent. This requires $H \geq P$ and holds generically when a non-linearity follows. The rank of M is then the number of independent windows, so the update reveals the length of the utterance.

2.3. Closed form and capacity

Let $\Pi^\perp$ be the projector onto $\mathcal{S}^\perp$, from the SVD of M . The true features satisfy

$$\Pi^\perp [c_p(X), 1] = 0, \quad p = 1, \dots, P. \quad (2)$$

Each c_p copies frames of X , so (2) is linear in X . Its normal matrix is block-banded with bandwidth kF and is solved by banded Cholesky in $O(TF(kF)^2)$. There are $P(kF + 1 - P)$ equations for about sPF unknowns, which gives the necessary condition

$$P \leq (k - s)F + 1. \quad (3)$$

Without overlap ($k = s$) each window may be any element of $\mathcal{S}$ and nothing is identified. The rank fixes P unless frames repeat exactly; we therefore take the shortest length whose solution lies in the span, and test uniqueness by perturbing the solve.

2.4. What federated training changes

After K local SGD steps the server forms $(W_0 - W_K)/\eta = \sum_j \sum_p g_p^{(j)} c_p^\top$. The windows do not depend on the weights, so the row space is unchanged. Momentum reweights the terms and weight decay adds a known multiple of W_0 . A batch places the windows of all its utterances in one span, so (3) bounds the total number of frames. Norm clipping rescales M . Dropout after the first layer alters g_p only. Per-coordinate optimisers such as Adam destroy the structure.

2.5. Noise

For an update $M + E$ with i.i.d. Gaussian E of standard deviation σ , $\|E\|_2 \approx \sigma(\sqrt{H} + \sqrt{kF} + 1)$ and Wedin's theorem bounds the rotation of the span by $\rho = \|E\|_2/s_P(M)$, where s_P is the weakest signal singular value. The error should grow linearly with ρ and the attack should fail near $\rho = 1$.

3. Beyond the first layer

Past the capacity the first-layer span is the whole space. Deeper layers still leak. In DeepSpeech-1 the second and third layers and the LSTM input weights act on $h_1(c_t)$, $h_2(c_t)$, $h_3(c_t)$, clipped-ReLU functions of a single window with 2,048 to 4,096 units. By (1) their gradients leak $\mathcal{S}_\ell = \text{span}\{[h_\ell(c_t), 1]\}$, of dimension T up to the layer width. The nonlinearity is essential: a linear stack would leak the same $(kF+1)$ -dimensional span at every layer.

A candidate window c is tested by the residual $r(c) = \sum_\ell \|\Pi_\ell^\perp[h_\ell(c), 1]\|^2 / \|[h_\ell(c), 1]\|^2$, which vanishes on the windows of the utterance and, generically, only there. The test does not say which frame a window is, and descending on all frames at once collapses to an average window. The zero padding provides the order. The first window is $[0, \dots, 0, x_1, \dots, x_{c+1}]$; we solve for it by Levenberg–Marquardt from the opening frames of public utterances. Each later window shares all frames but one with its predecessor, leaving F unknowns per step. We use the spans of the second and third layers; the LSTM-input span is noisier on trained weights and is needed only beyond 37 s. The last 26 frames are re-fitted jointly every 8 steps, which keeps the recursion from drifting. Decoding stops when c consecutive frames are zero: the padding at the far end. The length is thus an output. A joint refinement of all frames follows.

The same idea applies to any front end whose features are local in time. For Conv2dSubsampling followed by a linear projection [21], the projection leaks the span of 7-frame receptive fields, and a decoder can seed on any window and extend in both directions.

4. Experiments

4.1. Setup

Data: LibriSpeech [22] test-clean; one float32 gradient of the CTC loss per utterance. **DeepSpeech-1**: 26 MFCCs at 50 Hz, $c = 6$, three clipped-ReLU layers of 2,048 (4,096) units, a BiLSTM, two output layers. **Transformer-CTC**: 80 log-mel bins at 100 Hz, per-utterance normalisation, Conv1d(80→1536) with $k=7$, $s=3$ and a gated linear unit, 12 pre-norm blocks of width 768. Both are attacked at random initialisation and along training on train-clean-100. The attacker receives the update and the model. Public data (train-clean-100, disjoint speakers) supplies starting points and trains the vocoder front end. An utterance

Table 1: One update per utterance. SNR is the median over utterances of the feature-domain SNR. [†]length given (83% without). [‡]transcript and length given.

Target	Attack	n	Recov.	SNR
DS1, ≤ 6.2 s	closed form	1,417	98.5%	115
DS1, 6.3–35.0 s	sequential	191	96.9%	127
DS1 trained	closed form	60	98.3%	
DS1 trained, dropout 0.2	closed form	60	98.3%	
Transf., ≤ 9.6 s	closed form	600	94.8%	63
Transf. trained	closed form [†]	60	90%	
Gradient matching [‡] [14]	2,000 steps	4	0%	<3

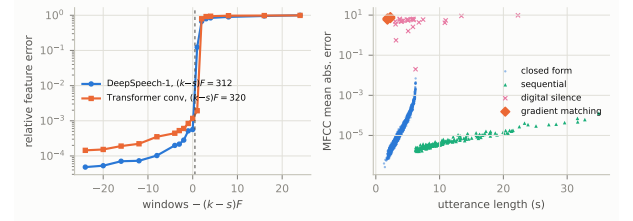


Figure 1: Left: relative error of the closed form around the predicted capacity. Right: MAE against utterance length for all attacked utterances.

counts as recovered if its mean absolute error (MAE) is below 10^{-2} for MFCCs, whose mean magnitude is 8, or if its SNR exceeds 30 dB for normalised log-mel.

4.2. Closed form

On DeepSpeech-1 the closed form recovers 98.5% of the 1,417 test-clean utterances that fit in the first layer (Table 1), with median MAE 1.4×10^{-5} in 2.5 s, and reads the length from the rank (Fig. 2). All 21 failures are flagged by the uniqueness test: the recordings contain runs of bit-identical frames (digital silence) that make windows identical. On the Transformer front end, 94.8% of 600 utterances up to 9.6 s are recovered at 63 dB. Fig. 1 (left) crops real features to exact lengths: the error stays below 10^{-3} up to $(k-s)F$ windows and reaches order one within two windows, for $(k, s, F) = (13, 1, 26)$, $(19, 1, 26)$, $(7, 3, 80)$ and $(5, 2, 80)$.

4.3. Long-form speech

The sequential decoder recovers 96.9% of 191 utterances between 6.3 and 35.0 s, with median MAE 4.0×10^{-6} , including the longest utterance of the set (1,748 frames). The remaining failures are digital silence. Decoding takes 37 s per utterance (median, one GPU) and involves neither the LSTM nor the loss. Checkpoints trained to 41% WER are decoded equally well (12 of 12 utterances).

4.4. Comparison with gradient matching

We rerun output-layer gradient matching through a twice-differentiable CTC with transcript and length given. After 2,000 Adam steps (2,973 s) on 2 s utterances the cosine distance is below 10^{-2} and the MAE is 5.9 to 8.0: the output-layer gradient has rank at most 29 and cannot determine hundreds of frames. Matching all layers exceeded 34 GB of GPU memory at 100 frames.

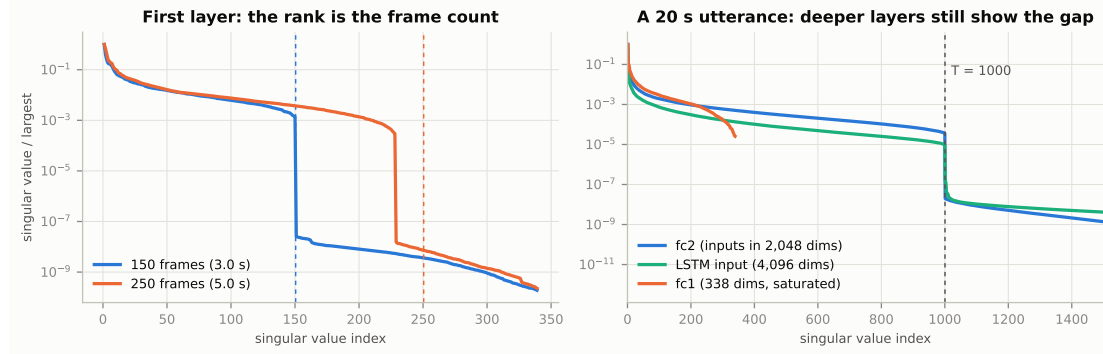


Figure 2: Singular values of gradients from DeepSpeech-1, relative to the largest. Left: first layer, two short utterances; the spectrum drops to the float32 floor exactly at the frame count. Right: a 20 s utterance saturates the first layer, but the second layer and the LSTM input weights still show the gap.

Table 2: Closed form on the first-layer update of DeepSpeech-1 under federated conditions (top, 20 trials each, update formed from float32 weight differences) and defences (bottom, 40 to 60 utterances each). Recovered: MAE below 0.1 (top), below 0.01 (bottom).

Condition	Median MAE	Recov. (%)
1 SGD step	$4.5 \cdot 10^{-3}$	85
10 SGD steps	$4.4 \cdot 10^{-3}$	100
50 SGD steps	$5.4 \cdot 10^{-3}$	90
SGD, momentum 0.9	$1.0 \cdot 10^{-4}$	95
SGD, weight decay	$5.2 \cdot 10^{-4}$	100
batch of 2	$1.6 \cdot 10^{-3}$	100
batch of 8 (short)	$1.3 \cdot 10^{-3}$	80
batch of 16 (short)	$1.1 \cdot 10^{-3}$	85
10 Adam steps	$7.4 \cdot 10^0$	0
dropout 0.2	$5.5 \cdot 10^{-6}$	98
dropout 0.5	$2.3 \cdot 10^{-6}$	98
clipping to 0.01	$1.2 \cdot 10^{-5}$	100
noise 10^{-5} of RMS	$7.1 \cdot 10^{-4}$	82
noise 10^{-3} of RMS	$6.6 \cdot 10^{-1}$	0
8-bit quantisation	$6.1 \cdot 10^0$	0
prune 50%	$6.8 \cdot 10^0$	0
sign only	$7.6 \cdot 10^0$	0

Table 3: Audio synthesised from recovered features: Whisper WER and speaker identification top-1.

Reconstructions	n	WER (%)		Speaker (%)	
		orig.	recov.	orig.	recov.
DS1, ≤ 6.2 s	197	3.0	3.8	99	98
DS1, 6.3–35 s	57	3.4	4.2	100	100
Transformer, ≤ 9.6 s	186	3.8	4.2	99	99

Table 4: Gradient rank of early layers on one 5 s utterance (Whisper and wav2vec 2.0 with released weights). A layer leaks a usable span when the rank equals the number of windows and is below both of its dimensions.

Model	Layer	Windows	Rank	Leaks
wav2vec 2.0	first conv	15999	10/10	no
wav2vec 2.0	feature proj.	249	249/512	yes
Whisper	conv1	3000	241/241	no
DeepSpeech-1	fc1	250	250/339	yes
Transf.-CTC	conv	166	167/561	yes
DeepSpeech-2	first conv	20331	32/32	no
DeepSpeech-2	RNN input	251	242/800	no
QuartzNet	depthwise conv	249	33/33	no
QuartzNet	pointwise conv	249	65/65	no
Conformer	first conv	—	10/10	no
Conformer	proj. after subs.	123	123/256	yes

4.5. Federated conditions and training

Table 2 forms the update from float32 weight differences. The closed form is unaffected by 1 to 50 local SGD steps, momentum 0.9 and weight decay (median MAE 10^{-4} to $6 \cdot 10^{-3}$; the floor is rounding in the subtraction and falls with larger steps). Batches of 2 to 16 utterances within capacity are recovered when the attacker knows their lengths; a batch over capacity is not. Dropout rates from 0.1 to 0.5 leave 98% recovered, in contrast with the protection reported against optimisation attacks [7]. DeepSpeech-1 checkpoints from initialisation to 41% WER are recovered at 98.3%. On the trained Transformer a float32 convolution artefact lies close to the weakest signal direction and the window count is sometimes off by one; with the length given, 90% are recovered.

4.6. From features to audio

A 6-layer Transformer maps recovered features to the mel input of a public HiFi-GAN [23]. Whisper small.en [24] transcribes

the audio and ECAPA-TDNN embeddings [25] assign it to one of 40 enrolled test speakers (Table 3). Since recovery is exact, audio from recovered and from true features scores the same: what the front end keeps, the update gives away.

4.7. What stops the attack

Local Adam, magnitude pruning of half the entries and sign compression break the low-rank structure (0% recovered). Gaussian noise behaves as predicted: the relative error equals about 0.5ρ up to $\rho \approx 0.3$, and the attack fails near $\rho = 1$, a noise level near 10^{-3} of the gradient RMS for utterances of a few seconds. Central differential privacy adds noise after aggregation and does not protect a client from the server.

Architecture matters most (Table 4). The first layers of pre-trained Whisper (inputs padded to 3,000 frames) and wav2vec 2.0 [26] (10-sample windows), of DeepSpeech-2 [27] (32 chan-

nels) and of QuartzNet [28] (depthwise) are saturated and leak no usable span. The wav2vec 2.0 feature projection and the projection after Conformer subsampling have rank equal to the frame count; on the latter our local decoder finds true windows but recovers only 9% of the frames on average (16 dB) before its chains stall, which we count as a negative result for now. Dropout stops the sequential decoder as well, since deeper layers see the dropped activations; it does not stop the closed form.

5. Conclusion

A wide linear layer over overlapping feature windows makes one federated update equivalent to the utterance, up to a capacity we can state in advance. Dropout, clipping, local SGD and training do not change this; breaking the low-rank structure or exceeding the capacity does. Limitations: the attack recovers features, and pitch in the audio is inferred by the vocoder; digital silence makes solutions non-unique; updates aggregated over many clients exceed the capacity. Code, per-utterance results and audio are in the supplementary material.

6. Generative AI Use Disclosure

A generative AI coding assistant was used to implement and run experiments and to draft this manuscript from the experimental results. The authors are responsible for the content and must revise the text before submission so that it complies with the Interspeech policy on generative AI.

7. References

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